

9.6 KEY TERMS, REVIEW QUESTIONS, AND PROBLEMS

Key Terms

arrival rate dispatcher exponential averaging fair-share scheduling fairness first-come-first-served first-in-first-out long-term scheduler	medium-term scheduler multilevel feedback predictability residence time response time round robin scheduling priority service time	short-term scheduler throughput time slicing turnaround time utilization waiting time
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Review Questions

- 9.1. Briefly describe the three types of processor scheduling.
- 9.2. What is usually the critical performance requirement in an interactive operating system?
- 9.3. What is the difference between turnaround time and response time?
- 9.4. For process scheduling, does a low-priority value represent a low priority or a high priority?
- 9.5. What is the difference between preemptive and nonpreemptive scheduling?
- 9.6. Briefly define FCFS scheduling.
- 9.7. Briefly define round-robin scheduling.
- 9.8. Briefly define shortest-process-next scheduling.
- 9.9. Briefly define shortest-remaining-time scheduling.
- 9.10. Briefly define highest-response-ratio-next scheduling.
- 9.11. Briefly define feedback scheduling.

Problems

- 9.1. Consider the following workload:

Process	Burst Time	Priority	Arrival Time
P1	50 ms	4	0 ms
P2	20 ms	1	20 ms
P3	100 ms	3	40 ms
P4	40 ms	2	60 ms

- a. Show the schedule using shortest remaining time, nonpreemptive priority (a smaller priority number implies higher priority) and round robin with quantum 30 ms. Use time scale diagram as shown below for the FCFS example to show the schedule for each requested scheduling policy.

Example for FCFS (1 unit = 10 ms):

P1	P1	P1	P1	P1	P2	P2	P3	P3	P3	P3	P3	P3	P3	P3	P3	P3	P4	P4	P4	P4
0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20

b. What is the average waiting time of the above scheduling policies?

9.2. Consider the following set of processes:

Process	Arrival Time	Processing Time
A	0	3
B	1	5
C	3	2
D	9	5
E	12	5

Perform the same analysis as depicted in Table 9.5 and Figure 9.5 for this set.

- 9.3. Prove that, among nonpreemptive scheduling algorithms, SPN provides the minimum average waiting time for a batch of jobs that arrive at the same time. Assume that the scheduler must always execute a task if one is available.
- 9.4. Assume the following burst-time pattern for a process: 6, 4, 6, 4, 13, 13, 13, and assume that the initial guess is 10. Produce a plot similar to those of Figure 9.9.
- 9.5. Consider the following pair of equations as an alternative to Equation (9.3):

$$S_{n+1} = \alpha T_n + (1 - \alpha) S_n$$

$$X_{n+1} = \min [Ubound, \max [Lbound, (\beta S_{n+1})]]$$

where *Ubound* and *Lbound* are prechosen upper and lower bounds on the estimated value of T . The value of X_{n+1} is used in the shortest-process-next algorithm, instead of the value of S_{n+1} . What functions do α and β perform, and what is the effect of higher and lower values on each?

- 9.6. In the bottom example in Figure 9.5, process A runs for two time units before control is passed to process B. Another plausible scenario would be that A runs for three time units before control is passed to process B. What policy differences in the feedback-scheduling algorithm would account for the two different scenarios?
- 9.7. In a nonpreemptive uniprocessor system, the ready queue contains three jobs at time t immediately after the completion of a job. These jobs arrived at times t_1 , t_2 , and t_3 with estimated execution times of r_1 , r_2 , and r_3 , respectively. Figure 9.18 shows the linear increase of their response ratios over time. Use this example to find a variant of response ratio scheduling, known as minimax response ratio scheduling, that minimizes the maximum response ratio for a given batch of jobs ignoring further arrivals. (*Hint*: Decide, first, which job to schedule as the last one.)
- 9.8. Prove that the minimax response ratio algorithm of the preceding problem minimizes the maximum response ratio for a given batch of jobs. (*Hint*: Focus attention on the job that will achieve the highest response ratio and all jobs executed before it. Consider the same subset of jobs scheduled in any other order and observe the response ratio of the job that is executed as the last one among them. Notice that this subset may now be mixed with other jobs from the total set.)
- 9.9. Define residence time T_r as the average total time a process spends waiting and being served. Show that for FIFO, with mean service time T_s , we have $T_r = T_s / (1 - \rho)$, where ρ is utilization.

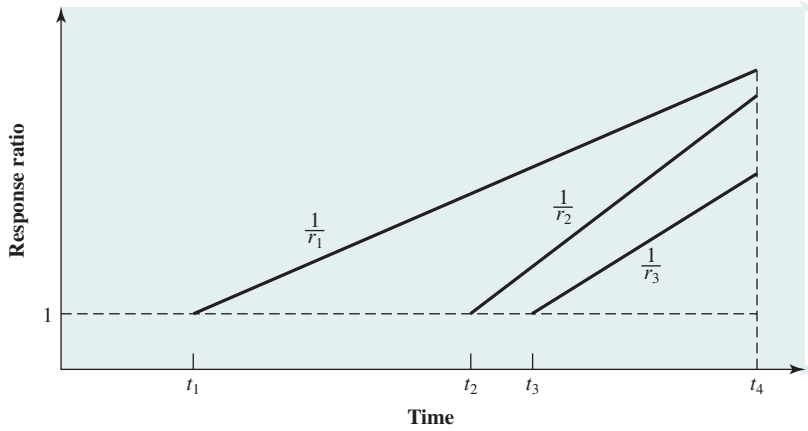


Figure 9.18 Response Ratio as a Function of Time

- 9.10.** A processor is multiplexed at infinite speed among all processes present in a ready queue with no overhead. (This is an idealized model of round-robin scheduling among ready processes using time slices that are very small compared to the mean service time.) Show that for Poisson input from an infinite source with exponential service times, the mean response time R_x of a process with service time x is given by $R_x = x/(1 - \rho)$. (*Hint:* Review the basic queueing equations in Appendix H or Chapter 20. Then consider the number of items waiting, w , in the system upon arrival of the given process.)
- 9.11.** Consider a variant of the RR scheduling algorithm where the entries in the ready queue are pointers to the PCBs.
- What would be the effect of putting two pointers to the same process in the ready queue?
 - What would be the major advantage of this scheme?
 - How could you modify the basic RR algorithm to achieve the same effect without the duplicate pointers?
- 9.12.** In a queueing system, new jobs must wait for a while before being served. While a job waits, its priority increases linearly with time from zero at a rate α . A job waits until its priority reaches the priority of the jobs in service; then, it begins to share the processor equally with other jobs in service using round robin while its priority continues to increase at a slower rate β . The algorithm is referred to as selfish round robin, because the jobs in service try (in vain) to monopolize the processor by increasing their priority continuously. Use Figure 9.19 to show that the mean response time R_x for a job of service time x is given by:

$$R_x = \frac{s}{1 - \rho} + \frac{x - s}{1 - \rho'}$$

where

$$\rho = \lambda s \quad \rho' = \rho \left(1 - \frac{\beta}{\alpha} \right) \quad 0 \leq \beta < \alpha$$

assuming that arrival and service times are exponentially distributed with means $1/\lambda$ and s , respectively. (*Hint:* Consider the total system and the two subsystems separately.)

- 9.13.** An interactive system using round-robin scheduling and swapping tries to give guaranteed response to trivial requests as follows: After completing a round-robin cycle among all ready processes, the system determines the time slice to allocate to each

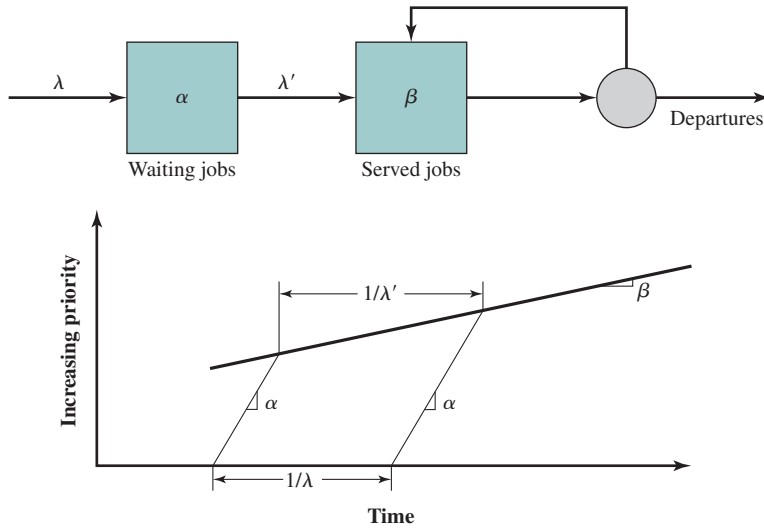


Figure 9.19 Selfish Round Robin

ready process for the next cycle by dividing a maximum response time by the number of processes requiring service. Is this a reasonable policy?

- 9.14.** Which type of process is generally favored by a multilevel feedback queueing scheduler—a processor-bound process or an I/O-bound process? Briefly explain why.
- 9.15.** In priority-based process scheduling, the scheduler only gives control to a particular process if no other process of higher priority is currently in the Ready state. Assume that no other information is used in making the process scheduling decision. Also assume that process priorities are established at process creation time and do not change. In a system operating with such assumptions, why would using Dekker's solution (see Section A.1) to the mutual exclusion problem be "dangerous"? Explain this by telling what undesired event could occur and how it could occur.
- 9.16.** Five batch jobs, A through E, arrive at a computer center at essentially the same time. They have an estimated running time of 15, 9, 3, 6, and 12 minutes, respectively. Their (externally defined) priorities are 6, 3, 7, 9, and 4, respectively, with a lower value corresponding to a higher priority. For each of the following scheduling algorithms, determine the turnaround time for each process and the average turnaround for all jobs. Ignore process switching overhead. Explain how you arrived at your answers. In the last three cases, assume that only one job at a time runs until it finishes and that all jobs are completely processor bound.
 - a.** round robin with a time quantum of 1 minute
 - b.** priority scheduling
 - c.** FCFS (run in order 15, 9, 3, 6, and 12)
 - d.** shortest job first